

Dear Editor,

Please consider the attached manuscript "Estimating the duration of RT-PCR positivity for SARS-CoV-2 from doubly interval censored data with undetected infections" for publication in the Biometric Practice section of Biometrics. The manuscript addresses a critical methodological challenge that arose during the COVID-19 pandemic: accurately estimating the duration of RT-PCR positivity from community prevalence surveys. This quantity is essential for estimating incidence of infection - key to informing pandemic response decisions - from prevalence data.

The work provides the first estimate of SARS-CoV-2 infection duration in a general population sample from validated statistical methods, providing substantial new insight beyond existing studies. To do this, we use Bayesian methodology to combine information from two studies with complementary strengths: one has high-frequency sampling but short-term follow-up and the other has low-frequency sampling but long-term follow-up.

Our work primarily uses data from the UK's Coronavirus Infection Survey (CIS), which was a globally unique, large-scale longitudinal study spanning almost three years of the pandemic. The study comprised a representative sample of randomly selected households, with consenting residents in each given a predetermined testing schedule. The aim was to estimate infection prevalence while removing the observational biases commonly associated with pandemic data. However, there are several statistical challenges associated with the CIS dataset.

To address the statistical challenges, we extend existing survival analysis frameworks to incorporate doubly interval censored data with imperfect test sensitivity and undetected episodes. While methods for each of these existed separately, their combination required a novel method. The methodology we developed is validated through simulation studies and sensitivity analysis, including an examination of the resilience to model misspecification. Our method is generic, allowing it to be applied to studies in future pandemics. Our simulation set up could be modified to address study design questions for future similar studies, providing a template that could be valuable for analysis of similar longitudinal studies in future public health emergencies.

We have made available all code to facilitate reproducibility and future applications of our approach. My PhD thesis (DOI: [10.17863/CAM.112028](https://doi.org/10.17863/CAM.112028)) contains the results and methods in this paper; this manuscript's text and figures have been heavily edited with additional information added, such as information on the participant's characteristics.

We feel that Biometrics is an ideal place to publish this research with its focus on an innovative methodological solution to a real-world data challenge, and we hope you agree.

Yours sincerely,

Joshua Blake